



Unit - I : -Fundamentals of Data Science



Dr. Gaikwad K. P. (Ph.D. CSE, M.Tech CSE, LMISTE)
Associate Professor
Artificial Intelligence and Data Science, MMIT, Pune

Contents

- **Basic Concepts of Data:** *Types of data:* Structured, Semi-structured, Unstructured, *Scales of measurement:* Nominal, Ordinal, Interval, Ratio, *Data formats:* CSV, JSON, XML, SQL tables, *Data quality, dimensions:* Accuracy, Completeness, Consistency, Timeliness, *Difference Between Data, Information, Knowledge, wisdom.*
- **Definition of Data Science:** *The Data Science Process/Lifecycle* (e.g., CRISP-DM, Team Data Science Process (TDSP), SEMMA Methodology), *Applications of Data Science in various domains* (e.g., Healthcare, Finance, Retail), *Distinction between Data Science, Machine Learning, and Artificial Intelligence, Roles in Data Science* (e.g., Data Analyst, Data Scientist, Machine Learning Engineer),
- **Ethical and Privacy Issues in Data Science.**

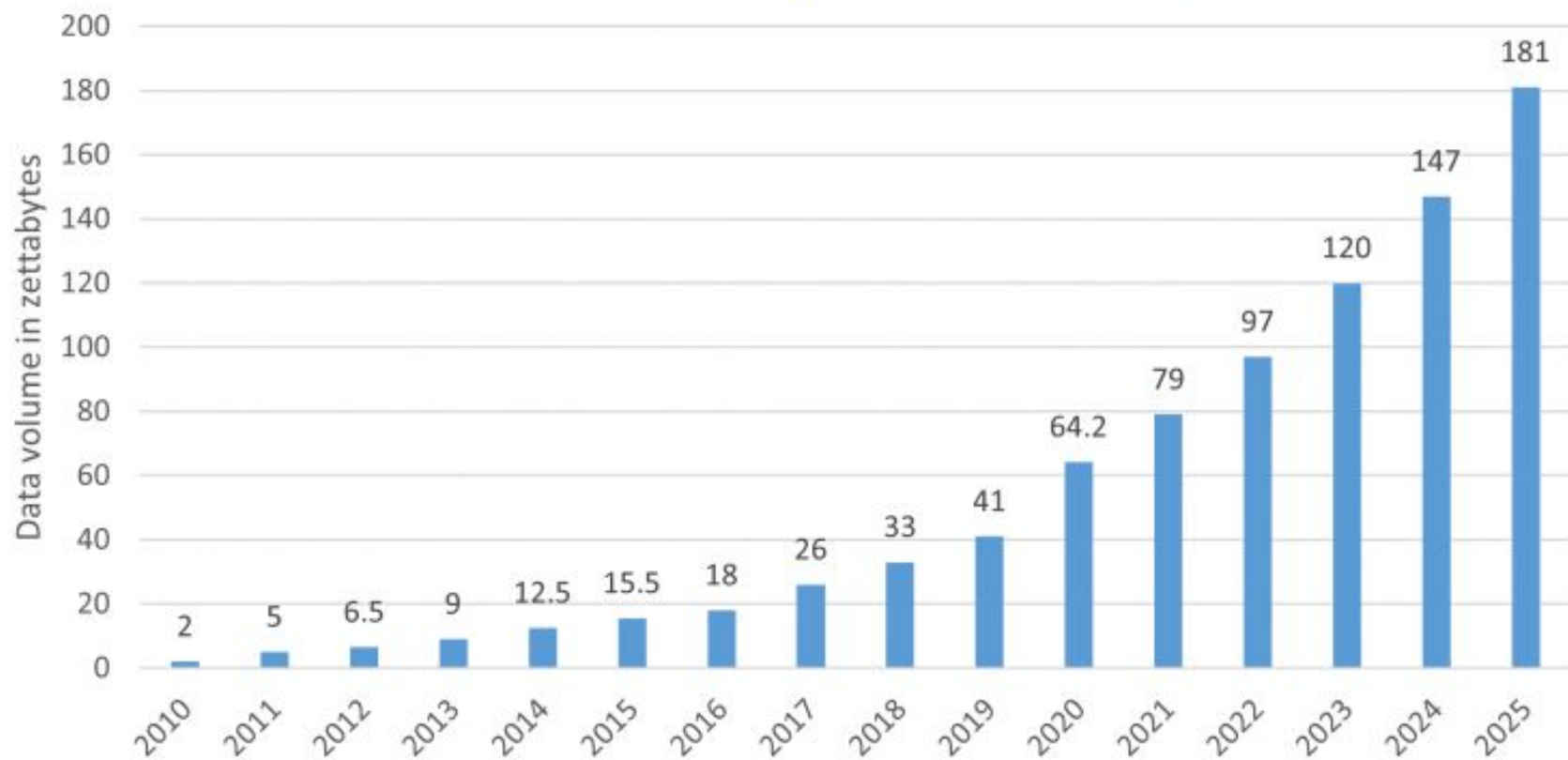
- ***Science*** is a discipline that studies and observes the natural world to understand how things work. It aims to build knowledge and make testable predictions about natural phenomena by formulating and testing hypotheses.
 - **Examples:** Biology, Chemistry, Physics, Microbiology, Geology, Astronomy, Neurobiology, Physiology
- ***Engineering*** is the application of scientific and mathematical principles to design, build, and maintain structures, machines, and systems. It involves solving practical problems by creating efficient and innovative solutions through the use of technology and materials.
 - **Examples:** Mechanical , Electrical, Civil, Chemical, Aerospace, Biomedical, Environmental, Computer

Part-I Basic Concepts of Data

Introduction to Data

- Data is **raw, unorganized facts, figures, symbols, or observations (like numbers, words, images, sounds)** that can be **collected, measured, and processed to extract meaningful insights, forming the basis for information and decision-making**, often stored digitally in bits and bytes.
- It represents realities or ideas in a formal way, but without organization, it lacks inherent meaning; processing it turns data into valuable information.
- The volume of data created, captured, copied, and consumed globally has grown exponentially over the last 15 years, increasing from an estimated 2 zettabytes (ZB) in 2010 to approximately 147 ZB in 2024, driven by widespread internet access, the rise of IoT devices, social media and AI.

Volume of data created and replicated worldwide (source: IDC)



What is Structured Data?

- **Structured data**, is the most organized form of data, designed for easy storage, access, and analysis.
- This data type is typically formatted into predefined rows and columns, making it highly searchable and easily organized within databases or spreadsheets.
- Each element in structured data is addressable, which means it can be precisely defined and easily grouped or related to other elements.
- Typically, structured data is housed in relational database management systems, which allows for complex querying and analysis using SQL.

Structured data

ID	Name	Age	Degree
1	John	18	B.Sc.
2	David	31	Ph.D.
3	Robert	51	Ph.D.
4	Rick	26	M.Sc.
5	Michael	19	B.Sc.

Examples of Structured Data

- **SQL Databases** (e.g., tables, rows, columns, primary keys, foreign keys)
- **Financial transactions** (e.g., sales data, purchase orders, accounting entries)
- **Customer demographic information** (e.g., name, address, age, gender)
- **Machine logs**, like events captured by devices, formatted with time stamps and specific parameters
- **Smartphone location data**, such as GPS coordinates captured at fixed intervals
- **Spreadsheets** that are commonly used for various business operations, from inventory to employee tracking

Examples of Structured Data



Pricing Data



CRM Data



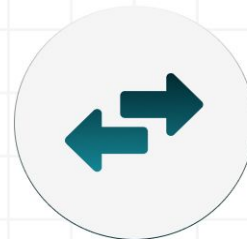
Dates & Times



**Customer
Account Data**



**Medical
Information**



**Financial
Transactions**

Advantages / Disadvantages of Structured Data

- **Easy analysis:** Because it is highly organized, structured data can be easily analyzed using standard tools like SQL queries.
- **Accuracy and consistency:** Fixed data fields reduce the chance of errors and provide uniformity.
- **Performance:** Structured data is optimized for relational databases, making searches and computations fast and efficient.
- **Limited flexibility:** Structured data requires a predefined schema, which means it can only store information that fits into a rigid format of rows and columns. This makes it difficult to accommodate dynamic or complex data.
- **Not suitable for all data types:** Real-world data is often complex and doesn't always fit neatly into structured fields. Structured data is not ideal for capturing qualitative information like images, videos, or long text, which can limit its applicability in certain areas.
- **Requires upfront planning:** Because unstructured data requires a well-defined schema, it often necessitates upfront design and planning, which can slow down agile projects or processes that involve rapid changes.

What is Unstructured Data?

- **Unstructured data** represents the largest category of data, and it's growing exponentially as more digital content is created.
- Unlike structured data, unstructured data doesn't follow a specific format or schema, which makes it more challenging to store and analyze.
- This type of data comes from a wide variety of sources, including emails, social media content, and multimedia files.
- Due to its lack of structure, unstructured data cannot be stored in traditional row-column databases and often requires more advanced storage solutions like data lakes.

Unstructured data

The university has 5600 students.

John's ID is number 1, he is 18 years old and already holds a B.Sc. degree.

David's ID is number 2, he is 31 years old and holds a Ph.D. degree. Robert's ID is number 3, he is 51 years old and also holds the same degree as David, a Ph.D. degree.

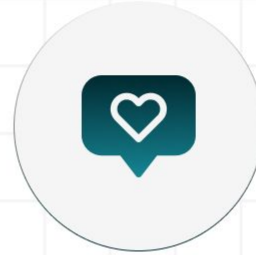
Examples of Unstructured Data



**Text Files
& Documents**



**Server, Website
& Applications Logs**



**Social
Media Data**



Sensor Data



Image Files



Audio Files



Video Files



Emails

Examples of Unstructured Data

- **Emails**—while certain fields, like the sender and timestamp, are structured, the email body itself is unstructured text.
- **Photos and videos**, because multimedia files are usually stored as raw data and lack predefined fields.
- **Audio files** (e.g., recordings of customer service calls, podcasts, and music files)
- **Text documents** (e.g., PDFs, Word documents, and open-ended survey responses)
- **Social media content**, such as posts, tweets, comments, and other user-generated content, all of which are unstructured and vary widely in format.
- **Call center transcripts or recordings**, while voice interactions can be analyzed for sentiment or trends, are naturally unstructured.

Advantages / Disadvantages of Unstructured Data

- **Rich in insights:** Unstructured data, especially from sources like social media or customer feedback, often contains nuanced and valuable information.
- **Flexibility:** Unstructured data can capture complex, real-world scenarios that structured data cannot.
- **Sentiment analysis and brand identification:** AI algorithms can analyze unstructured data for patterns, trends, and sentiments that structured data may not reveal.
- **Versatility:** With tools like AI and ML, unstructured data can now be harnessed for applications such as predictive maintenance (from machine logs) and fraud detection.
- **Difficult to store and manage:** Traditional databases cannot handle unstructured data, meaning organizations must invest in alternative storage solutions like data lakes, which require specialized management.
- **Challenging to analyze:** Extracting useful insights from unstructured data is more difficult and often requires sophisticated tools like AI and ML, which may not be readily available to all organizations.
- **Resource-intensive:** Processing and analyzing unstructured data can require more computational power, specialized software, and skilled personnel. This makes extracting value from it more costly and time-consuming than extracting value from structured data.
- **Quality and consistency issues:** Unstructured data is often inconsistent in format and quality, making it harder to standardize and ensure accuracy during analysis. The lack of uniformity in unstructured data can lead to unreliable insights if not processed carefully.

What is Semi-Structured Data?

- **Semi-structured data** blends elements of both structured and unstructured data.
- It doesn't fit neatly into relational databases, it contains some organizational markers, such as metadata or tags, that make it easier to manage and analyze than fully unstructured data.
- Semi-structured data typically doesn't follow rigid schemas but has some structure that tools can leverage to provide useful insights.
- In that sense, semi-structured data strikes a balance between structured data's rigidity and unstructured data's flexibility, making it a valuable format for modern businesses looking to work with complex data sources.

Semi-structured data

```
<University>  
  <Student ID="1">  
    <Name>John</Name>  
    <Age>18</Age>  
    <Degree>B.Sc.</Degree>  
  </Student>  
  <Student ID="2">  
    <Name>David</Name>  
    <Age>31</Age>  
    <Degree>Ph.D. </Degree>  
  </Student>  
  ....  
</University>
```

Examples of Semi-Structured Data



01

JSON Files



02

XML Documents



03

CSV & Log Files



04

HTML Documents



05

Emails



06

YAML Documents

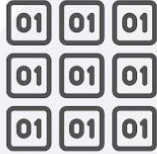
Examples of Semi-Structured Data

- Web technologies (e.g., HTML)
- NoSQL databases (e.g., MongoDB, CouchDB, CockroachDB)
- DevOps (e.g., log files)
- JSON, XML, and YAML, which are common formats for semi-structured data that contain tags and elements but are not rigidly organized like relational databases.

Advantages / Disadvantages of Semi-structured Data

- **Flexibility with organization:** You can store large volumes of data with some structure, making it easier to analyze than fully unstructured data.
- **Ideal for web and IoT data:** Many modern data formats, such as those used in web apps or IoT devices, are semi-structured, making them more versatile.
- **Supports scalability:** Scaling semi-structured data storage solutions can be easier than traditional relational databases.
- **Less efficient than structured data:** While semi-structured data offers more flexibility, it is still less efficient to store and query than fully structured data, which is optimized for fast, complex queries.
- **Requires specialized tools:** Semi-structured data can't be easily handled by traditional relational databases, requiring organizations to adopt more specialized tools like NoSQL databases or specific analytics platforms.
- **Consistency is harder to ensure:** Because semi-structured data doesn't adhere to strict schemas, it can be challenging to maintain consistency across datasets, especially as they grow in size and complexity.
- **Limited standardization:** Unlike structured data, semi-structured data doesn't have industry-wide standardization, which can lead to compatibility issues when integrating with other systems or platforms.

Structured data



Characteristics

Predefined data models
Easy to search
Text-based
Shows what's happening

Resides in

Relational databases
Data warehouses

Stored in

Rows and columns

Examples

Dates, phone numbers, social security numbers, customer names, transaction info

Unstructured data



Characteristics

No predefined data models
Difficult to search
Text, pdf, images, video
Shows the why

Resides in

Applications
Data warehouses and lakes

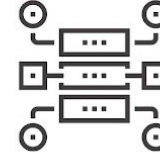
Stored in

Various forms

Examples

Documents, emails and messages, conversation transcripts, image files, open-ended survey answers

Semi-structured data



Characteristics

Loosely organized
Meta-level structure that can contain unstructured data
HTML, XML, JSON

Resides in

Relational databases
Tagged-text format

Stored in

Abstracts & figures

Examples

Server logs, tweets organized by hashtags, emails sorting by folders (inbox; sent; draft)

Scales of Measurement

Nominal

Ordinal

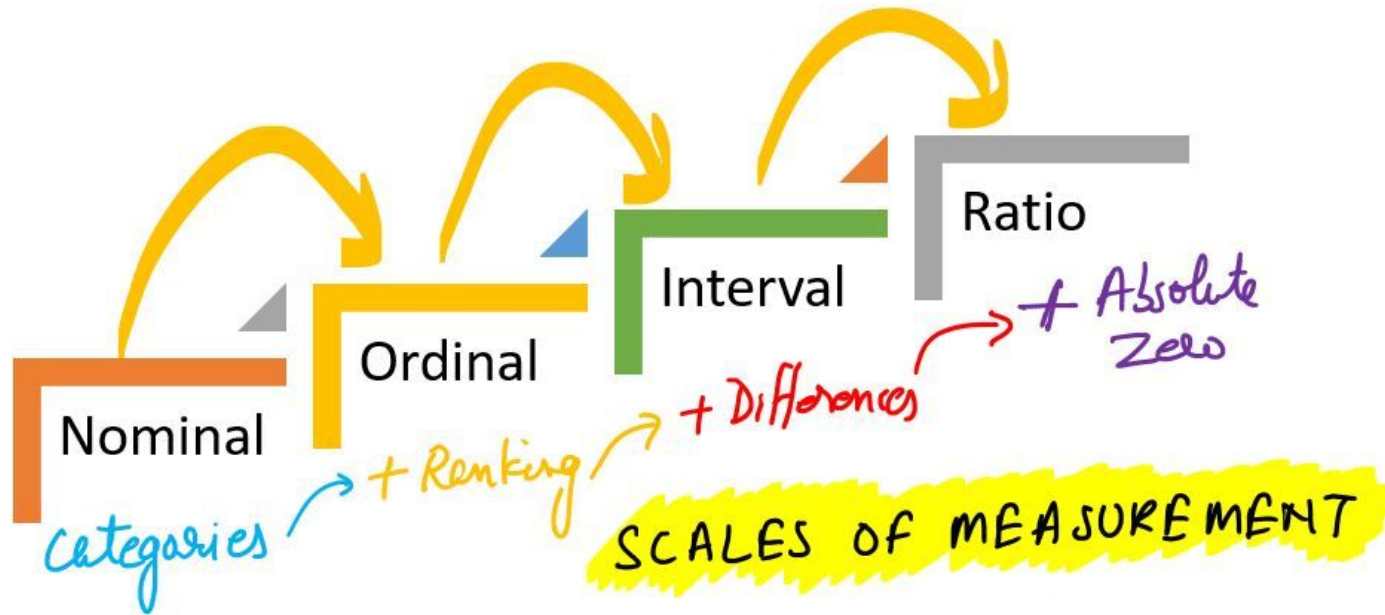
Interval

Ratio



What is the Scale of Measurement?

- **Scale of Measurement:** Scale of measurement refers to the ways variables or data are classified and quantified.
- There are four main levels — **Nominal, Ordinal, Interval, and Ratio** — each with increasing levels of precision and mathematical usefulness.
 - **Nominal:** the data can only be categorized
 - **Ordinal:** the data can be categorized and ranked
 - **Interval:** the data can be categorized, ranked, and evenly spaced
 - **Ratio:** the data can be categorized, ranked, evenly spaced, and has a natural zero.



Nominal Scale of Measurement

- **Nominal Scale**, also called the categorical variable scale, is defined as a scale that labels variables into distinct classifications and doesn't involve a quantitative value or order.
- There are cases where this scale is used for the purpose of classification – the numbers associated with variables of this scale are only tags for categorization or division.
- **Example:** Where do you live?
 - 1- Suburbs
 - 2- City
 - 3- Town
- A nominal scale is often used in research surveys and questionnaires where only variable labels hold significance.
 - A customer survey asking “Which brand of smartphones do you prefer?” Options: “Apple”- 1, “Samsung”-2, “OnePlus”-3.

Nominal Scale Data and Analysis

There are two primary ways in which nominal scale data can be collected:

1. By asking an *open-ended question*, the answers of which can be coded to a respective number of labels decided by the researcher.
2. The other alternative to collect nominal data is to include a *multiple-choice question* in which the answers will be labeled.

What is your Gender?
• M- Male • F- Female

What is your Political preference?
• 1- Independent • 2- Democrat • 3- Republican

Ordinal Scale of Measurement

- **Ordinal Scale** is defined as a variable measurement scale used to simply depict the order of variables and not the difference between each variable. These scales generally depict non-mathematical ideas such as frequency, satisfaction, happiness, a degree of pain, etc.
- **Ordinal Scale** maintains descriptive qualities along with an intrinsic order but is void of an origin of scale, and thus, the distance between variables can't be calculated. Descriptive qualities indicate tagging properties similar to the nominal scale, in addition to which the ordinal scale also has a relative position of variables.
- **Ordinal Data and Analysis**
 - **Ordinal scale** data can be presented in tabular or graphical formats for a researcher to conduct a convenient analysis of collected data.

Ordinal Scale Examples-1

- Status at the workplace, tournament team rankings, order of product quality, and order of agreement or satisfaction are some of the most common examples of the ordinal Scale.
- These scales are generally used in market research to gather and evaluate relative feedback about product satisfaction, changing perceptions with product upgrades, etc.
- Example: How satisfied are you with our services?
 - Very Unsatisfied – 1
 - Unsatisfied – 2
 - Neutral – 3
 - Satisfied – 4
 - Very Satisfied – 5

Ordinal Scale Examples-2

1. Here, the order of variables is of prime importance, and so is the labeling. Very unsatisfied will always be worse than unsatisfied, and satisfied will be worse than very satisfied.
2. This is where the ordinal scale is a step above the nominal scale – the order is relevant to the results, and so is their naming.
3. Analyzing results based on the order along with the name becomes a convenient process for the researcher.
4. If they intend to obtain more information than what they would collect using a nominal scale, they can use the ordinal scale.

Interval Scale of Measurement

- Interval Scale is defined as a numerical scale where the variables' order is known and the difference between these variables.
- These scales are effective as they open doors for the statistical analysis of provided data. Mean, median, or mode can be used to calculate the central tendency in this scale.
- The only drawback of this scale is that there is no pre-decided starting point or a true zero value.
- The interval scale contains all the properties of the ordinal scale and offers a calculation of the difference between variables. The main characteristic of this scale is the equidistant difference between objects.
- A Celsius/Fahrenheit temperature scale –
 - Eighty degrees is always higher than 50 degrees, and the difference between these two temperatures is the same as the difference between 70 degrees and 40 degrees.
 - Also, the value of 0 is arbitrary because negative temperature values exist – which makes the Celsius/Fahrenheit temperature scale a classic example of an interval scale.

Interval Scale Applications

- **Psychological Testing:** IQ scores and other standardized tests.
- **Weather Analysis:** Temperature comparison in Celsius or Fahrenheit.
- **Educational Research:** Measuring attitudes using Likert scales.
 - a. A *Likert scale* is a popular survey tool where respondents rate their level of agreement or disagreement with statements, typically on a 5 or 7-point scale (e.g., Strongly Agree to Strongly Disagree). Developed by psychologist *Rensis Likert*, it quantifies attitudes, opinions, and perceptions.

Ratio Scale of Measurement

- **Ratio Scale** is defined as a variable measurement scale that not only produces the order of variables but also makes the difference between variables known, along with information on the value of true zero.
- Because of the existence of a true zero value, the ratio scale doesn't have negative values.
- To decide when to use a ratio scale, the researcher must observe whether the variables have all the characteristics of an interval scale along with the presence of the absolute zero value.

Ratio Scale Applications -1

- What is your daughter's current height?
 - Less than 5 feet.
 - 5 feet 1 inch – 5 feet 5 inches
 - 5 feet 6 inches- 6 feet
 - More than 6 feet
- What is your weight in kilograms?
 - Less than 50 kilograms
 - 51- 70 kilograms
 - 71- 90 kilograms
 - 91-110 kilograms
 - More than 110 kilograms

Ratio Scale Applications -2

- **Scientific Research:** Measuring time, weight, length, or volume.
- **Economics & Finance:** Income levels, expenditure, market prices.
- **Health & Fitness:** Tracking body weight, heart rate, or exercise duration.



Data Formats

- In the world of data analytics, the choice of data format plays a crucial role in efficiency, storage, and processing.
- Different formats cater to various needs, from simple text-based exchanges to optimized binary storage for big data systems.



CSV (Comma Separated Values)

- CSV is a straightforward text format where each row of data is a line, and values within the row are separated by commas (or other delimiters). It's like a basic spreadsheet without any fancy features.
- CSV is popular because it's easy to generate, read, and compatible with most tools, but it lacks built-in schema or data types, which can lead to parsing issues.

```
Name, Age, City  
John, 30, New York  
Alice, 25, Los Angeles  
Bob, 35, Chicago
```

e_id	e_name	e_salary	e_age	e_gender	e_dept
1	Sam	95000	45	Male	Operations
2	Bob	80000	21	Male	Support
3	Anne	125000	25	Female	Analytics
4	Julia	73000	30	Female	Analytics
5	Matt	159000	33	Male	Sales
6	Jeff	112000	27	Male	Operations

SQL (Relational Table Format)

- SQL represents data in relational tables, which are like grids with rows (records) and columns (fields).
- It's not a file format itself but a way to structure data in databases.
- Each table has a defined schema specifying data types, and you can query it using SQL language.
- It's great for structured data with relationships but requires a database system to manage.

```
CREATE TABLE students (  
    name VARCHAR(255),  
    register_number INT,  
    subject VARCHAR(255),  
    marks INT  
);  
  
INSERT INTO students (name, register_number, subject, marks) VALUES ('Alice', 101,  
'Math', 90);  
INSERT INTO students (name, register_number, subject, marks) VALUES ('Bob', 102,  
'Science', 85);  
INSERT INTO students (name, register_number, subject, marks) VALUES ('Charlie', 103,  
'English', 95);
```

JSON (JavaScript Object Notation)

- JSON is a flexible, text-based format that stores data as key-value pairs (objects) or lists (arrays).
- It's human-readable, supports nested structures, and is widely used in web services, APIs, and configuration files.

```
{
  "orders": [
    {
      "orderno": "748745375",
      "date": "June 30, 2088 1:54:23 AM",
      "trackingno": "TN0039291",
      "custid": "11045",
      "customer": [
        {
          "custid": "11045",
          "fname": "Sue",
          "lname": "Hatfield",
          "address": "1409 Silver Street",
          "city": "Ashland",
          "state": "NE",
          "zip": "68003"
        }
      ]
    }
  ]
}
```

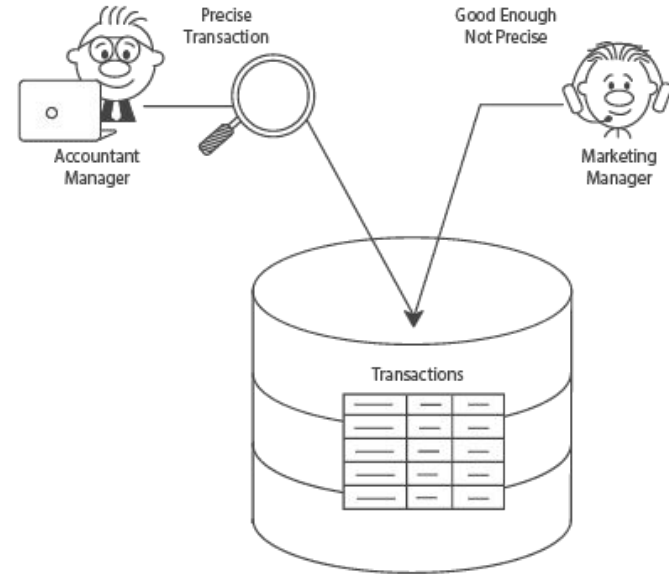

XML (Extensible Markup Language)

- XML is a text-based markup language that uses hierarchical tags to structure data. It's like a tree of elements, making it suitable for complex, nested data.
- XML is verbose and self-descriptive but less efficient for large volumes due to its size. It's common in enterprise systems and web services.

```
<?xml version="1.0" encoding="UTF-8"?>
<catalog>
  <book id="001">
    <title>Harry Potter and the Sorcerer's Stone</title>
    <author>J.K. Rowling</author>
    <genre>Fantasy</genre>
    <price>20.00</price>
    <publish_date>1997-06-26</publish_date>
  </book>
  <book id="002">
    <title>The Great Gatsby</title>
    <author>F. Scott Fitzgerald</author>
    <genre>Classic</genre>
    <price>15.00</price>
    <publish_date>1925-04-10</publish_date>
  </book>
</catalog>
```


What is Data Quality?

- **Data Quality (DQ)** is defined as the data's suitability for a user's defined purpose. It is subjective, as the concept of quality is relative to the standards defined by the end user's expectations.
- **Data Quality Expectations:** It is possible that for the exact same data, different users may have totally different data quality expectations, depending on their usage.
 - For example, the **accounting department** needs accurate data to the salary, whereas the **marketing team** does not, because the approximate sales numbers are enough to determine the sales trends.



Data Quality Dimensions

Uniqueness

Prevents duplicate entries for the same entity.



Accuracy

Ensures data correctly reflects real-world entities.



Validity

Verifies data conforms to defined standards.



Completeness

Guarantees all necessary data is available.



Timeliness

Ensures data is current and available when needed.

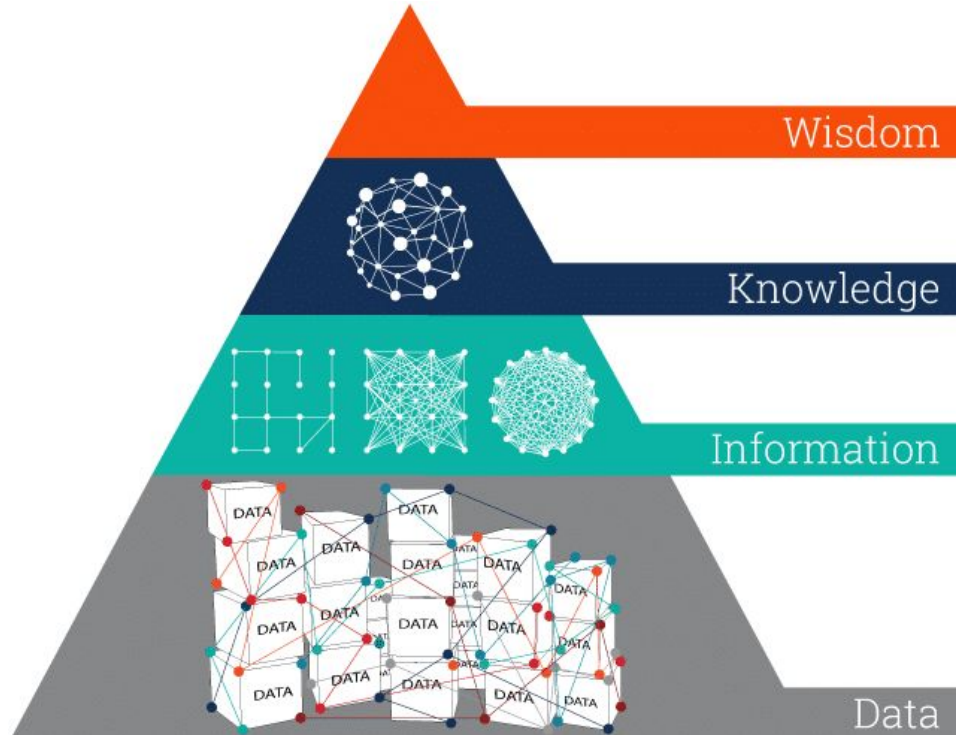


Consistency

Maintains uniformity across different data systems.



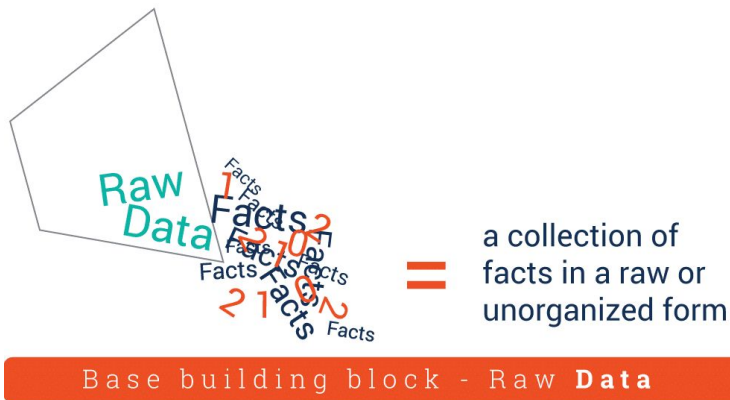
Data, Information, Knowledge, Wisdom (DIKW) Pyramid



- The DIKW Pyramid represents the *relationships between data, information, knowledge and wisdom*

Data

- **Data** is a collection of facts in a raw or unorganized form such as numbers or characters.
- However, **without context, data can mean little.**
 - For example, *12012012* is just a sequence of numbers without apparent importance.
 - But if we view it in the context of ‘this is a date’, we can easily recognize *12th of January, 2012*.
- **By adding context and value to the numbers, they now have more meaning.**



Information

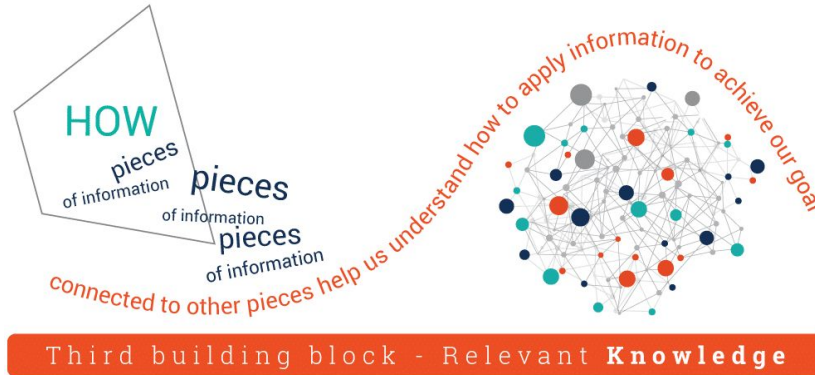
- **Information** is the next building block of the DIKW Pyramid. This is data that has been “cleaned” of errors and further processed in a way that makes it easier to measure, visualize and analyze for a specific purpose.
- By asking relevant questions about ‘who’, ‘what’, ‘when’, ‘where’, etc., we can derive valuable **information from the data** and make it more useful for us.
- But when we get to the question of ‘how’, this is what makes the leap from information to Knowledge.



Second building block - Derived **Information**

Knowledge

- When we don't just view information as a description of collected facts, but also understand how to apply it to achieve our goals, we turn it into knowledge.
- “How” is the information, derived from the collected data, relevant to our goals? “How” are the pieces of this information connected to other pieces to add more meaning and value? And, maybe most importantly, “how” can we apply the information to achieve our goal?
- **But only when we use the knowledge and insights gained from the information to take proactive decisions**, we can say that we have reached the final – ‘wisdom’ – step of the Knowledge Pyramid.



Wisdom

- Wisdom is the top of the DIKW hierarchy and to get there, we must answer questions such as ‘why do something’ and ‘what is best’. In other words, wisdom is knowledge applied in action.
- We can also say that, if data and information are like a look back to the past, knowledge and wisdom are associated with what we do now and what we want to achieve in the future.



Examples on DKIW - 1

E-commerce (Amazon)

- **Data:** Amazon collects millions of clicks, purchases, and reviews.
- **Information:** Data is processed into customer profiles showing buying patterns (e.g., buys dog food, organic treats).
- **Knowledge:** Algorithms know that customers who buy these items often buy specific toys or grooming tools.
- **Wisdom:** Amazon recommends those specific toys to the customer, increasing sales and user satisfaction.

Examples on DKIW - 2

- **Fitness Tracking :** Fitness tracking devices collect your health and activity data, but your end goal is to use that to make decisions about how to train or how to manage your health.
 - **Data:** The smartwatch collects raw data such as the number of steps taken, heart rate, and sleep duration.
 - **Information:** The smartwatch app organizes and structures the data, displaying it in a comprehensible format, such as daily step count, average heart rate, and hours of sleep per night.
 - **Knowledge:** Analyzing and interpreting the information may reveal patterns, such as increased step count leading to improved sleep quality or a correlation between heart rate and workout intensity.
 - **Wisdom:** Understanding these patterns lets you make informed decisions about adjusting your exercise routine, sleep habits, and other lifestyle factors to improve your health and fitness.

Part-II Introduction to Data Science

What is Data Science

- “A field of study that uses scientific methods, processes, and systems to extract knowledge and insights from data”.
- **Data science** is inherently **interdisciplinary** as it combines expertise from **statistics, computer science, mathematics, and domain-specific knowledge**.
- This makes it incredibly versatile, with applications spanning healthcare, finance, marketing, and even environmental research.
- **Example - Predictions about poll results:** They aren't baseless guesses; they're insights from public opinion surveys, voter turnout models, and a variety of advanced tools and methodologies used in data science

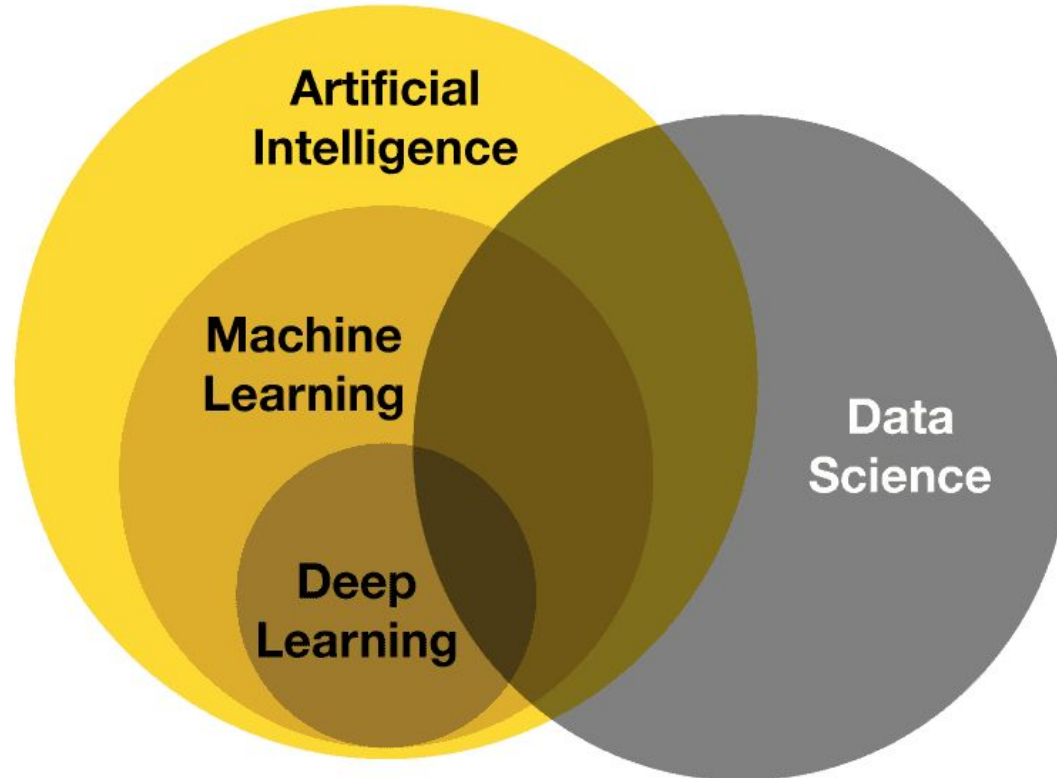


Statistics

[stə-'ti-stiks]

A branch of applied mathematics that involves the collection, description, analysis, and inference of conclusions from quantitative data.

Distinction between Data Science, Machine Learning, and Artificial Intelligence



Distinction between Data Science, Machine Learning, and Artificial Intelligence

Artificial Intelligence

- imparts human intellect to machines
- uses logic and decision trees
- includes machine learning



Machine Learning

- subset of AI
- uses statistical models
- machines improve with experience

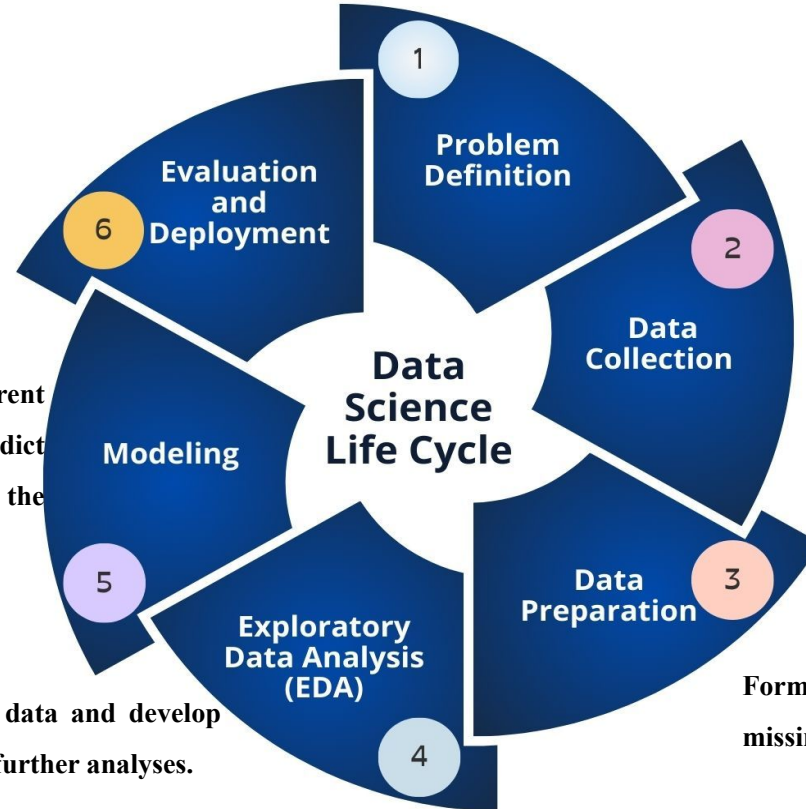


Data Science

- based on strict analytical evidence
- deals with structured & unstructured data
- includes various data operations



The Data Science Life Cycle



Collect data from a variety of sources, including databases, sensors, APIs (application programming interfaces), and online platforms

Format the data for analysis and deal with missing values, duplicates, and other errors

Use machine learning algorithms or different statistical techniques in order to predict outcomes or explain relationships within the data.

The goal is to better understand the data and develop hypotheses to guide model building or further analyses.

The Data Science Life Cycle

- **Obtaining data :** The first step in the data science life cycle is obtaining the data. At this stage, the most important thing is ensuring that you have the right information to work with.
- **Cleaning data:** After the data is collected, data scientists clean and preprocess it. This step, often referred to as data cleaning or wrangling
- **Exploring data:** At this stage, the data is ready, so data scientists begin the so-called exploratory data analysis (EDA) process. The aim is to understand the data's underlying structures and main characteristics and identify patterns. Depending on the number of variables analyzed at a time, EDA can utilize univariate, bivariate, or multivariate analysis.
- **Modeling data:** These models can be predictive, such as forecasting future sales, or descriptive, such as clustering customers by behavior.
- **Interpreting results:** If all previous steps are done correctly, data scientists should have produced results by the end of the cycle, and all that is left to do is interpret the conclusions and communicate them to the rest of the team. All insights should be presented in a clear and concise manner so that stakeholders can understand them and be able to use them to aid decision-making.

Use Cases in Finance



Use Cases in Healthcare



Data Science for
Medical Imaging



Data Science for
Drug Discovery



Data Science for
Genomics



Tracking and
Preventing Diseases



Predictive
Analytics



Data Science for
Wearables

Use Cases in Retail



Recommendation
engines



Price
optimization



Cross-selling and
upselling



Fraud
detection



Personalized
marketing

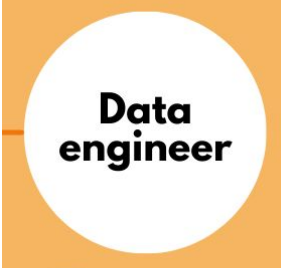


Customer sentiment
analysis

Roles in Data Science-1



**Data
scientist**



**Data
engineer**



BI analyst



**Data
analyst**



**ML
engineer**

•

Roles in Data Science-2

- A **Data Analyst** extracts, cleans, and organizes structured datasets to **uncover trends and generate dashboards or reports for stakeholders**. Provide descriptive analytics that inform immediate business decisions, such as sales performance or customer segmentation.
- A **Data Scientist** leverages programming, statistics, and machine learning frameworks (e.g., Scikit-learn, TensorFlow) to **build and validate models that predict future outcomes or prescribe actions**. Address complex questions — like forecasting demand or optimizing pricing — by developing predictive and prescriptive analytics
- A **Machine Learning Engineer** focuses on converting Data Science prototypes into production services, handling data pipelines, model deployment, monitoring, and CI/CD for ML systems. Maintain reliable, efficient, and reproducible ML operations in live environments.
- A **Data Engineer** designs, builds, and maintains the systems (data pipelines/infrastructure) that collect, store, and **process large volumes of raw data, transforming it into a reliable, accessible format for data scientists and analysts to derive insights**.
- A **BI (Business Intelligence) Analyst** gathers, cleans, analyzes, and visualizes data to help companies make smarter, data-driven decisions, transforming complex information into understandable reports and dashboards for executives and teams, using tools like SQL, Tableau, and Power BI.

Ethical and Privacy Issues in Data Science.



**Privacy
Protection**



**Responsible
Data Use**



**Informed
Consent**



**Security
Safeguards**



**Fairness
and Equity**



**Data
Integrity**



**Transparency
and
Explainability**



**Algorithmic
Accountability**



**Ethical
Decision-
making**

Case Study

- A telecom business wants to proactively identify customers who are likely to quit because of the high rate of customer turnover. The organization wants to predict loss of customers using data science techniques and take proactive steps to keep at-risk clients.
- Determine the steps that, as data scientists, we must adhere to in order to resolve this issue.